

A Hashtag Is Worth a Thousand Words: An Empirical Investigation of Social Media Strategies in Trademarking Hashtags

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Abstract. Firms of all sizes are “joining the conversation” on social media platforms and increasingly trademarking hashtags related to their products and brands. This added effort to protect intellectual property and its impact on social media engagement have not been investigated in the literature. In this study, we attempt to bridge this important gap in the literature by first examining the impact of trademarking a hashtag on a firm’s social media audience engagement. By adopting multiple causal identification strategies to address the issues of self-selected trademarking, we find that trademarking hashtags plays a pivotal role in increasing social media audience engagement and information dissemination. More importantly, this positive effect is stronger for firms with fewer Twitter followers. Digging deeper into the underlying mechanisms, we find that trademarking hashtags makes composing tweets with certain linguistic styles more critical: It can amplify the positive effects of trademarking hashtags on social media audience engagement. Our findings highlight important managerial implications of trademarking hashtags. First of all, we examine whether trademarking a hashtag helps or hurts a firm in terms of its social media audience engagement. Further, we show, to maximize the effectiveness of trademarking hashtags, how firms should develop the right social media engagement strategies by taking specific communication and linguistic styles into account. Our results provide useful insights to firms in understanding the key benefits of signaling through trademarking hashtags on social media engagement.

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We don’t have a choice on whether we do social media, the question is how well we do it.

—Erik Qualman, author of *Socialnomics*¹

We bet it all on the hashtag.

—Josh Decker, CEO and founder of startup Tagboard (O’Brien 2015)

1. Introduction

In the past decade, social media platforms have become an important channel by which consumers and firms communicate and engage with each other (Bapna et al. 2017, Kumar et al. 2018b, Hu et al. 2019, Kumar et al. 2019, Chung et al. 2020, Kumar and Qiu 2022). Engaging with consumers through social media helps businesses reduce information asymmetry, gain insight into their consumer bases, turn highly dissatisfied customers

into satisfied ones, and elevate satisfied consumers into loyal buyers, leading to an increase in sales (Kumar et al. 2018a) and brand profitability (Lee et al. 2018a).

To engage and motivate consumers to evaluate the firm’s products or services, firms usually employ different types of signals (Kwark et al. 2021, Hsu et al. 2022). With the help of these signals, consumers derive expectations about the quality of the firm’s characteristics that cannot be observed directly (Qiu et al. 2015, Lee et al. 2018b, Yang and Yuan 2019). An example of such signals to send credible information to consumers is a trademark (Chemmanur et al. 2020). Trademarks are typically used by firms that want to protect their intellectual properties and to differentiate their products or services from competitors (Morris et al. 2006), encourage branding efforts (Keller 1993), and amplify shareholder value (Davidson 2004). The increase in the

acceptance of trademarks has allowed them to be used as a standard for measuring a firm's innovation (Flikkema et al. 2019) and its future product development and marketing strategy (Faurel et al. 2021).

A trademark reflects a set of complex attributes or characteristics that may not be readily observed (Economides 1988, Swann et al. 2001). Trademarks typically undergo a strict scrutiny process established by the trademark office.² The trademark act requires a verified statement supporting the application signed by the owner or a person properly authorized to sign on behalf of the owner. The penalty imposed for an improper filing of intellectual property such as patent and trademark may far exceed that of merely losing it (MacKendrick et al. 2015, Philipp 2021, Siesel 2021). For example, it may formally exclude a party from bringing matters before the trademark office as an attorney, correspondent, domestic representative, and or signatory (USPTO 2021b).³ Therefore, firms have inherent incentives to provide accurate information to the trademark office and follow the process to avoid facing penalties. Because of the strict filing and auditing process, for firms holding trademarks, consumers are certain of receiving messages from authentic sources, which further provides some assurances that the signals conveyed by firms are reliable. In addition, using a trademark provides credible signals of a firm's main performance measures, such as enhancing shareholder value, seeking higher initial public offering pricing, increasing revenue and future profitability, and realizing higher stock returns (Davidson 2004, Hsu et al. 2022).

Although "using trademarks" has been widely studied, they mainly focus on the role of trademarking brands. Recently, firms have begun trademarking hashtags to protect this valuable form of intellectual property (Marshall 2016, Roberts 2017). Hashtags are typically used on social media as ubiquitous conversation starters, marketing campaign tools, and brand symbols. The benefits of using hashtags include boosting businesses' brand recognition, generating buzz around new products, and attracting new customers while ensuring a high level of engagement (Lewis 2016, Buckhanon 2019). Original hashtags give firms control over a social media space in which they can show off their industry expertise, drive viral conversations, and promote a special event, contest, or campaign to interested consumers (Blaszka et al. 2016, McKelvey and Grady 2017, Morales 2018). In this study, we examine a particular form of firm-side social media signaling strategy—trademarking a hashtag—which has not been examined in prior literature.

1.1. Motivation

The U.S. Patent and Trademark Office (USPTO) confirmed the registrability of hashtags as trademarks in 2013. The trademark is frequently rated as a primary

intangible property owned by a firm (Yang and Yuan 2019). An increasing number of firms have been trademarking hashtags and taking advantage of this extra protection (Davey 2016). Per the Trademark Manual of Examining Procedure (TMEP 2021), hashtags can become registered trademarks only if [they] function as an identifier of the applicant's goods and services.⁴ This registration allows firms to gain a competitive advantage and build reputational value on social media with increased distance between firms and consumers (Sherwin 2015).

To the best of our knowledge, little is known in the research community and practitioners' guidelines about the signaling effect of a trademarked hashtag on audience engagement in social media platforms. What is the effect of a trademarked hashtag on audience engagement in social media platforms? Using online platforms, consumers have limited capacity to analyze a product and its quality, as they do not have direct means to feel, touch, or inspect a product (Lal and Sarvary 1999, Mehra et al. 2018). These limitations lead to higher information asymmetry in online channels, making it more important for firms to send credible signals, such as trademarking, when compared with offline channels (Biswas and Biswas 2004). Once a firm registers a trademark, consumers may recognize specific goods and services or perceive differences among brands. Therefore, a trademark may reduce consumers' search costs by condensing complex meanings into concise, easy-to-remember, and unequivocal hashtags (Swann et al. 2001, Ty Inc. vs. Perryman 2002, Sipe 2021). In addition, because many consumers are aware of a trademark owned by a specific brand, exclusive rights in a trademark eventually may induce pressure on the firm to adhere to a consistent quality of the product or service (Katz 2010). It allows consumers to depend on the firm's trademark as a mental shortcut, thereby further reducing the overall search costs (Tushnet 2008). Therefore, a trademark helps firms make stronger bonds with consumers and further reduces information asymmetry.

Firms use various types of signals to influence consumers on social media (Mavlanova et al. 2012; Martin et al. 2015; Qiu and Whinston 2017; Qiu et al. 2018; Paruchuri et al. 2019; Na et al. 2020; Wang et al. 2021, 2022; Pu et al. 2022). Users adjust their perceptions of the firm depending on the type of signals. Signals revealed directly by the firm or individuals, without any impartial third-party endorsement or through unreliable sources, may not be perceived as credible signals by consumers (Mavlanova et al. 2012, Williams et al. 2015). On the other hand, the signals from impartial sources, such as the trademark office, are difficult, expensive, and time consuming to acquire (Connelly et al. 2011, Cheng et al. 2020). However, they are considered credible because of the strict filing and

auditing process established by the trademark office and have a more positive influence on consumers in forming their perception toward the firm and the way consumers react on social media (Winter and Krämer 2014). More specifically, firms with trademarks issued by the agency, such as the USPTO, are perceived to be sending more credible and authentic signals to consumers. These firms invest considerable resources to establish their brand equity, which serves as a credible signal to separate them from those who are unwilling to invest in their brand equity.

In our case, it is unclear whether the specific act of trademarking a hashtag plays the role of a credible signal on social media platforms where information asymmetry is usually high and perceiving direct product or service quality is difficult. Drawing on the signaling theory, a firm can send signals to convey its commitment to build and protect its brand equity to consumers as well as to clarify any consumer confusion about their brand (Hsu et al. 2022). This signaling, in turn, leads to an increase in the intentions of the consumers to engage with the firm and purchase its product or service (Jeng 2016). From a big-picture perspective, we fill this important research gap by examining the signaling effects of trademarking a hashtag.

In terms of precise motivation, trademarking hashtags with the USPTO may send credible signals to consumers, which may reduce information asymmetry on social media (Hsu et al. 2022). On the one hand, the act of signaling provides key insights into which a firm influences consumer behavior (Yang and Yuan 2019). More specifically, signaling through trademarking conveys vital information on a firm's new product, services, and marketing efforts (Faurel et al. 2021). The phenomenon of trademarking a hashtag may help in sending credible signals about firm commitment in building brand awareness, loyalty, and trust among consumers (Crass et al. 2019). Note that because the signaling effect depends on consumer awareness, in our context, firms publicize their acceptance of a trademarked hashtag on social media, such as Twitter, to ensure that the signal has been received by the consumers (see Figure D.1 in Online Appendix D). Therefore, a firm's act of trademarking may serve as a signal of a firm's credibility and increase audience engagement on social media. Moreover, the trademark protection of hashtags sends reliable signals to competitors and prevents them from using similar hashtags to mislead consumers (Chu 2017).

On the other hand, one could argue that a trademarked hashtag is not directly reported in a firm's financial statement or as a separate performance measure. Their economic value is also not straightforward to estimate (Seethamraju 2003). In addition, it is not easy to get virtually exclusive rights on a trademarked

hashtag because of the trademark dilution phenomenon, which makes it difficult to sue competitors for trademark infringement (Heath and Mace 2020). Therefore, trademarked hashtags may not help reduce the information asymmetry on social media because the audience may not know about the newly acquired intangible assets of these firms. Overall, the act of trademarking hashtags by firms may not send credible signals and may not have a significant effect on audience engagement on social media.

Given these two conflicting arguments concerning how trademarking a hashtag affects information asymmetry, our overarching theme is to empirically investigate whether a firm's act of trademarking a hashtag can lead to an increase in social media engagement via a signaling perspective. A careful empirical analysis will give researchers and practitioners the tools and evidence to aid firms in their ongoing attempts to maximize the effectiveness of their social media engagement strategies.

The information systems literature provides a high-level direction on content strategies associated with firm-generated contents (FGCs), information asymmetry, and consumer behavior and engagement. It includes broadcasting client-focused postings (Miller and Tucker 2013), delivering compelling content (Culnan et al. 2010), and providing quality content while fostering member embeddedness (Porter and Donthu 2008). This stream of literature has only recently begun to delve into what specific type of FGCs is associated with consumer engagement in brand communities (Lee et al. 2018a, Bapna et al. 2019). Overall, our study adds to the literature around FGCs, signaling theory, and information asymmetry phenomenon on social media.

Several studies in the past have examined the signaling role of third-party affiliations (Gulati and Higgins 2003). In addition, several other types of information technology-supported cues (such as technology or users-oriented features incorporated in online platforms/websites) are used by the firms as signaling sources to enhance user perception and experience (Cenfetelli et al. 2008, Li et al. 2015, Chen et al. 2017, Qiu et al. 2021). For example, improving website quality, displaying simple features, such as signs or a firm's logo, incorporating a firm's address, or interactive features (such as live chat) on the firm's online platform are all signaling sources (Wells et al. 2011, Mavlanova et al. 2012). However, none of these studies has looked into the signaling effect of trademarking a hashtag. Though the signals through these cues help consumers make positive perceptions about the firm and improve their experience, they may not be treated as authentic, reliable, or credible as trademarked hashtags as they do not require scrutiny from any credible and impartial sources. Our results show

that firms' intellectual assets on social media (i.e., trademarked hashtags) act as credible quality signals in social media engagement strategies.

Note that to understand the importance of trademarking hashtags, one should understand the difference between trademarks and copyrights, as they are commonly misunderstood and confused. Both forms of protection do not require registration. However, for the owner to have maximum legal weight, they must be registered. Copyrights protect the intellectual property created in a fixed or tangible form, such as books, arts, or recorded music.⁵ In most cases, names, titles, catchphrases, familiar symbols, methods, discoveries, and public domain cannot be copyrighted. Trademarks fill in the gaps where copyrights do not.

The common term trademarking also refers to another protection called a service mark. Trademarks are used in the protection of goods, whereas service marks protect services. Unlike copyrights, trademarks have the potential to never expire as long as the owner continues to pay annual fees. Note that the mark can only be considered if it "functions as an identifier of the source of the applicant's goods or services."⁶ This definition means if the hashtag contains words or a phrase that is too generic, it will likely be rejected for trademark protection (TMEP 2021). Based on the case *Eksouzian vs. Albanese*⁷ where the court deemed hashtags as descriptive devices rather than trademarks (and thus cannot be protected), one may argue that the actual legal protections are somewhat dubious. In our private conversation with Professor Alexandra J. Roberts, a Professor of Intellectual Property at the University of New Hampshire School of Law, she kindly provided her endorsement on the legal protection of trademarking hashtags as follows: "I would, of course, disagree with the assertion that 'actual legal protections are somewhat dubious.'" Her detailed quote is provided in Online Appendix E.

1.2. Research Questions

Although "using trademarks" has been widely studied, the literature has mainly focused on the role of trademarking brands. Past literature on trademarks provides a positive association of trademarks with a firm's degree of advancement regarding innovation (Schmoch and Gauch 2009), marketing (Mendonça et al. 2004), new product or service development (Helmers and Rogers 2011), productivity growth, and stock market value (Greenhalgh and Rogers 2006). The ability to build a sustainable brand on social media through trademarks can be one of the most crucial success factors for a firm (Aaker 2004). Also, trademarks assist firms in attaining a competitive advantage by differentiating their products or services from their competitors (Besen and Raskind 1991).

We examine a particular form of firm-side social media signaling strategy—trademarking hashtags—which has not been addressed in past research. Signaling theory provides a framework for understanding how extrinsic cues—signals—can be used by firms to convey service or product quality information to consumers, thereby reducing information asymmetry and driving purchase decisions (Wells et al. 2011). Our study draws on signaling theory and examines the signaling effect of trademarking a hashtag on information asymmetry measured through social media engagement. To the best of our knowledge, no prior research has quantified the signaling effects of trademarking a hashtag. To systematically bridge this research gap, we begin by asking our first research question: (1) *What is the signaling effect of trademarking a hashtag on a firm's social media audience engagement (i.e., effectiveness)?*

Drawing on the prior literature (Wells et al. 2011, Jeng 2016, Yang and Yuan 2019, Faurel et al. 2021, Hsu et al. 2022), we expect that a firm's act of trademarking a hashtag serves as a credible signal of a firm's seriousness and commitment on social media. The signaling effect of a trademarked hashtag helps reduce information asymmetry (increase audience engagement) by protecting consumers from confusion about social media promotions conducted by similar, competing companies. It also protects businesses by helping maintain the brand exclusivity they have worked hard to build. Therefore, we may observe a positive effect of trademarking a hashtag on a firm's social media audience engagement. More specifically, the act of trademarking a hashtag can significantly increase a firm's social media audience engagement (retweets). A careful empirical analysis will aid firms in their ongoing attempts to maximize the effectiveness of their social media engagement strategies. In addition, an empirical challenge of examining our first research question lies in establishing the causal effect of signaling through trademarking hashtags due to self-selection bias. Using several causal identifications strategies, our study is the first to provide an empirically driven, comprehensive understanding of trademarking hashtags.

Second, after comprehensively examining the signaling effect of trademarking a hashtag on a firm's social media audience engagement, it is natural to take a step further and investigate how the magnitude of the effects may differ among different types of firms. We investigate the heterogeneity of the signaling effect of trademarking hashtags in a social media context through the lens of the signaling theory. In general, the signaling effect of credibility carried by trademarking is more critical for one type of firm versus another (Cao and Hsu 2011). The strength of signals depends on the level of information asymmetry:

Signals have more informational value (the effect of signaling becomes larger) when the information asymmetry is higher (Connelly et al. 2011, Hottenrott et al. 2016). In our context, consumers might not be familiar with firms with fewer Twitter followers, including their products and services, because of high information asymmetry. Therefore, the positive effect of signaling through trademarking a hashtag may be larger for firms with fewer Twitter followers. Note that the number of Twitter followers may indicate a firm's efforts invested in social media. We use the number of Twitter followers to measure how consumers are familiar with the firm. If a firm has a large number of Twitter followers, it indicates that consumers are more familiar with the firm. The reason can be that the firm invests more in social media or has a larger size. Actually, firm size and social media efforts are strongly correlated. Prior studies find that the larger firms with larger market capital expend more effort on social media and engage in communication and gain popularity (Xiong et al. 2018, Xiong et al. 2019).

On the other hand, one could also argue that the signal strength of the positive effect of trademarking a hashtag may be larger for firms with more Twitter followers. For these firms, the signaling effect of trademarking a hashtag might be limited to reducing information asymmetry because these firms and their social media accounts are already well known (Yang and Yuan 2019). Both scenarios are theoretically plausible; the dominating effect may depend on the specific type of firm, thereby requiring empirical investigation. Therefore, we ask our second research question: (2) *Does the magnitude of the signaling effect of trademarking hashtags differ among firms with different levels of Twitter followers (i.e., heterogeneity)?*

We empirically find that the beneficial impact of trademarking a hashtag is not uniform across all types of firms, which suggests that the signaling effect of trademarking hashtags can be particularly important for certain types of firms. The positive impact of trademarking hashtags on social media engagement is only significant for firms with severe information asymmetry problems (e.g., firms with a fewer number of followers). Therefore, the effect of signaling through a trademark might be more important for these firms because consumers have limited knowledge (because of high information asymmetry) about them (Yang and Yuan 2019).

Next, we empirically investigate how a firm's use of social media to communicate with consumers impacts their social media effectiveness after an expected signaling event—trademarking a hashtag. It is important to understand not only what firms express, but the way in which firms express themselves on social media has some consequential (action-based) effects

on the audience. What writing style, based on psychological constructs (linguistic characteristics), of the FGCs prompts the audience to retweet after a firm's act of trademarking a hashtag? We examine this mechanism from the lens of speech act theory, which has been adopted in a wide variety of domains, such as information systems, consumer behavior, and marketing (Ludwig et al. 2016, Aleti et al. 2019, Villarroel et al. 2019, Fan and Ilk 2020).

The speech act theory deals with how words are used not only to present information but also to carry out actions (Austin 1975). The speech act, that is, the act of utterance of a word (FGC in our context), is typically divided into three categories: locutionary act, illocutionary act, and perlocutionary act. A locutionary act is an act of utterance that is used to simply express something. Illocution is a speech act where one presents intent through tone, attitude, feelings, or emotions to achieve something. Perlocution is a persuasion-based speech act that has some consequential effects on the audience. Through the lens of speech act theory, we assess the role of specific linguistic styles of firm-generated tweets after trademarking hashtags. We develop linguistic styles based on psychological constructs (using speech act theory) that are correlated with the perlocutionary effect on the audience, which, in turn, persuade the audience to further share information by retweeting. We then ask our third research question: (3) *What key linguistic characteristics (based on writing style) are associated with an increase in a firm's social media audience engagement after trademarking a hashtag?*

We draw on the prior literature and use the following key constructs: (i) Emotionally Charged FGCs, (ii) Narrative Style FGCs, and (iii) Inclusive and Confident Style FGCs. The usage of highly charged words in writing activates strong emotions, provokes consumers, and causes perlocution by sharing the contents frequently and widely (Rimé 2009). Drawing on the literature on the news advertising and persuasive effect of firms' interaction with consumers using a positive emotional tone (Berger and Milkman 2012, Araujo et al. 2015, Wang et al. 2021), we expect that emotionally charged words (positive) cause a perlocutionary effect (retweet) on social media. Narrative writing style involves the adoption of a storytelling style and using more descriptive words in writing. Consistent with the literature (Escalas 2007) on viral news content using a narrative style, we expect that the use of narrative writing style arouses excitement among consumers and subsequently persuades them to retweet. Also, with inclusiveness and confidence in writing style, the audience typically feels more empowered and knowledgeable and subsequently feels proud in sharing/diffusing the FGCs (Kim et al. 2016). Overall, we expect that the act of trademarking

a hashtag makes the frequency and writing style (Clout/Inclusive, Narrative, and Positive Style) of FGC more critical and can amplify the beneficial effects (perlocutionary effect) of writing more tweets with desirable linguistic styles.

Finally, linguistic styles of writing tweets may have both contemporaneous and lagged effects (Friedman 2011). It is important to examine the relationship between writing tweets with desirable linguistic features and the duration its effect lasts. The effect of writing more tweets with desirable linguistic styles may or may not be short-lived. Therefore, we ask our fourth research question: (4) *What are the contemporaneous and lagged relationships between writing tweets with desirable linguistic styles and social media audience engagement after trademarking hashtags?* The answer to this question provides insights into how long the beneficial effect of specific engagement strategies on social media may last after hashtags have been trademarked.

We expect that before trademarking hashtags, the effect of writing more tweets with desirable linguistic styles is short-lived; only the contemporaneous effect is significant (Friedman 2011). One month later, the lagged effect may no longer be significant. On the other hand, after trademarking hashtags, we see a correlation between writing more tweets with desirable linguistic styles and a longer duration of its effect. Therefore, corresponding to our research questions 3 and 4, we expect that after trademarking a hashtag, the amplified effects of writing tweets with desirable linguistic features may not only be more intense (answer to research question 3) but may also last longer (answer to research question 4).

1.3. Key Contributions

Our paper contributes to the literature on signaling theory, FGCs, and social strategy by showing the positive effect the act of trademarking a hashtag can have on social media audience engagement. We also highlight how the magnitude of the signaling (positive) strength varies by the number of followers associated with a firm because of information asymmetry. Our study also contributes to the speech act communication theory by testing how the adoption of specific linguistic styles (based on psychological construct) can have a long-term perlocutionary effect on consumer behavior, hence influencing them to further share information on social media through retweets.

With respect to contribution to practice, although it is costly to register, monitor, and renew a trademarked hashtag in the long run, the beneficial effects of trademarking can still exist and help firms promote and differentiate products from the competition on social media. If business decision makers do not understand the beneficial effects of trademarking

hashtags and do not have a good estimate about the magnitude of those effects, they will not invest sufficient resources (underinvestment) in trademarking and managing hashtags.

Note that firms may apply for a trademark on a hashtag according to their business intuitions in practice. However, the cost-benefit analysis of trademarking a hashtag is still in its infancy. In business practice, the benefit of a firm's credible signaling through trademarking hashtags is directly related to the causal effect on social media engagement and is difficult to estimate. Without estimating the magnitude of the causal effect of trademarking hashtags on social media engagement, the signaling effect of trademarking hashtags may not be optimal (overinvestment or underinvestment). Specifically, the efforts and subsequent benefits derived from using a trademarked hashtag might not be the same among different firms, especially firms with a smaller number of Twitter followers. Our findings have important implications for the social media strategies of certain types of firms. As part of managerial implications, our study provides guidelines, through the lens of signaling theory and speech act theory, on how firms generate FGCs and communicate on social media become even more relevant after trademarking hashtags. Also, we provide insights into not only how social media engagement strategies can amplify social media effectiveness after hashtags have been trademarked but also how long the beneficial effects may last. In summary, our study helps advance understanding of the benefits of trademarking hashtags and answers questions that are important to a wide range of stakeholders.

2. Literature Review

Our work is related to the following streams of literature: (i) Social media signaling and its impact on consumer behavior, (ii) content development and sharing strategies in social media, and (iii) hashtags and trademarking. In this section, we briefly review each stream of literature and highlight our contributions. Our main contributions to literature are summarized in Figure D.2 in Online Appendix D.

2.1. Social Media Signaling and Its Impact on Consumer Behavior

There are mainly two different ways by which contents are generated and consumed by consumers and firms on social media: user-generated content (UGC) and firm-generated content (Kumar et al. 2016, Qiu and Kumar 2017, Kumar and Qiu 2022). UGC represents consumer-to-consumer communication or consumer-to-business communication. Consumer participation in a firm's social media effort increases the intensity of the relationship between the firm and its

customers and leads to an increase in the frequency of customer visits (Rishika et al. 2013). As extensively documented in the literature, many studies have focused on the effect of UGC on key metrics, including sales (Rui et al. 2013, Dewan and Ramaprasad 2014) and user engagement, such as content diffusion on social media (Susarla et al. 2012, Qiu et al. 2015). By participating in the firm-initiated social media site, customers can voice their opinions and share experiences about the firm with other customers as well as the firm itself (Agarwal et al. 2009). Rishika et al. (2013) examine consumer transaction data with UGC on social media. Using a firm's Facebook page, Goh et al. (2013) discover a positive influence of firm-generated data and user-generated data on consumer purchase expenditures. Our study differs from this stream of research by focusing on the impact of FGC on the effectiveness of a firm's social media signaling strategy.

Firms are increasingly engaging with customers on social media by generating FGC, and its use on social media platforms is important because it influences how consumers view the firm. Firms regularly interact with their customers through social media profiles, such as Facebook pages and Twitter accounts. Their use of Twitter to disseminate information has been widely reported in the literature. Chen et al. (2015) examine how firm-generated messages drive additional sales through a change in consumer behavior. He et al. (2017) explore firms' social media strategies in product-harm crises. Specifically, they study how nonfocal firms adjust their joint social media strategies (offensive and defensive marketing) after a major unexpected incident. Ma et al. (2015) find that firm interaction with customer compliments and complaints on Twitter results in improved customer relationships but at the same time encourages more complaints later. Kumar et al. (2016) investigate the effect of FGC on social media on consumer behavior and spending. Chung et al. (2020) examine the impact of firm-generated social media responses to different types of consumer messages on the market performance of the firms. Lee et al. (2018b) study the relationship between different types of firm-generated advertising content and consumer behavior, such as the number of likes and comments.

Increasing engagement with customers via FGC can lead to enhanced customer loyalty and trust (Algesheimer et al. 2005) as well as better interactions and deeper connections between consumers and firms. A more active and vibrant community with regular messaging creates trust and allows customers to infer the level of a firm's commitment to bolstering the customer-firm relationship. Among different engagement measures, we follow Kumar et al. (2010), Oestreicher-Singer and Zalmanson (2013), Shi et al. (2014), Mallipeddi et al. (2021), and Mallipeddi et al. (2022) and use the number

of retweets (content sharing) as a performance metric for social media audience engagement. Retweeting reflects the active participation critical for businesses seeking to engage consumers as the information disseminated in each retweet exposes a new or larger audience to the firm's content. Kumar et al. (2010) point out that sharing a brand's social media content creates engagement and amplifies the impact of the original content. Shi et al. (2014) also argue that content sharing (retweeting) is a critical mechanism of information diffusion in social broadcasting networks and is vital to a network's proper functioning and thriving. According to Oestreicher-Singer and Zalmanson's (2013, p. 591) "ladder of participation" paradigm, content sharing is a very high level of online community participation and endorsement. Marketing practitioners also acknowledge that retweeting intensity is an appropriate measure of audience engagement on Twitter (Long 2015).

We extend the prior literature on FGC and consumer behavior to understand the impact of trademarking hashtags on consumer engagement in online brand communities. More specifically, we focus on one specific type of firm-side social media signaling strategy, trademarking hashtags. This has not been addressed in past research and is very important because an increasing number of firms have been applying for trademarks on hashtags (Davey 2016). Prior literature has investigated the role of trademarks as a signal to protect intellectual properties, differentiate their products, promote brands, strengthen shareholder value, showcase future product releases, and realize higher stock returns in various ways (Keller 1993, Davidson 2004, Morrin et al. 2006, Flikkema et al. 2019, Faurel et al. 2021, Hsu et al. 2022). Our work aims to contribute to the information systems literature by extending the literature on signaling theory (in social media) and consumer behavior by examining the effect of trademarking a hashtag on social media engagement (retweets). Because signals have a stronger effect when the information asymmetry is higher (Connelly et al. 2011, Hottenrott et al. 2016, Khurana et al. 2019), we further examine the heterogeneity effect, that is, whether the signaling strength of trademarking a hashtag differs among firms with different levels of Twitter followers because of the information asymmetry phenomenon. A careful empirical analysis will give researchers and practitioners the tools and evidence to aid firms in their ongoing attempts to maximize the effectiveness of their social media engagement strategies.

2.2. Content Development and Sharing Strategies in Social Media

Prior information systems literature provides a high-level direction on content strategies associated with engagement, including client-focused postings (Miller and Tucker 2013), a continuous supply of compelling content (Culnan et al. 2010), and providing quality

content and fostering member embeddedness (Porter and Donthu 2008). This stream of literature has only recently begun to delve into what specific type of content is associated with consumer behavior and engagement in brand communities. Lee et al. (2018a) find that posts related to brand personality (e.g., humor, emotional appeals, casual banter, philanthropic efforts) have a positive effect on consumer behavior, whereas informative content (that can be used to optimize purchasing decisions, for example, price, availability, promotions, and product facts) by itself is associated with lower levels of engagement. An essential question in this research stream is what types of firm posts are associated with engagement in online brand communities.

Our paper contributes to the literature by investigating the role of trademarking hashtags: in particular, how firm posts with a trademarked hashtag are associated with engagement in online brand communities. Prior literature (e.g., Ransbotham and Galbraith 2010) suggests that firms use posts as a means of influencing the behavior of others. One way in which firms can introduce this influence is through the use of speech act communications theory. We explain the key drivers or speech act style (psychology-based linguistic characteristics) that cause trademarking hashtags to increase a firm's audience engagement through the lens of speech act theory (Austin 1975) and linguistic features (Lee et al. 2015) in social media.

Prior studies in information systems, consumer behavior, and marketing have relied on the emerging literature on the application of speech act theory and psychology-based linguistic features in social media (Ludwig et al. 2016, Aleti et al. 2019, Villarroel et al. 2019, Fan and Ilk 2020). Lee et al. (2015) develop linguistics-based, quality-level indicators to assess the content quality of user-generated textual content. They find that the use of certain linguistic features, such as pronouns and other function words, is correlated with the content's perceived quality. Several studies have used linguistics-based features to detect deception (fake reviews) on social media. Ludwig et al. (2016) use texts of archived email exchanges to derive linguistic cues and detect deception in computer-mediated communication. Ho et al. (2016) create spontaneous deception scenarios in an interactive online game environment and identify language-action cues (e.g., cognitive load, affective process, latency, and wordiness) that reveal patterns of information behavior manifested by deceivers in online communication. To the best of our knowledge, using psychology-based linguistic features and speech act theory to examine the impact of trademarking a hashtag is a new and important area of research and has not been explored in the prior literature.

2.3. Hashtags and Trademarking

The hashtag, also known as the “pound sign,” has revolutionized the way information is organized, discovered, and shared online. The idea of using “#” as an indicator for groups of posts was formally adopted by Twitter in 2009. Since then, the hashtag's use began to grow, exploding in 2011, the year Instagram started hyperlinking hashtags. Fifty percent of all Super Bowl commercials contained a hashtag by 2013; in the same year, Facebook supported the use of hashtags via hyperlinks (Kolowich 2014). Hashtags are most prevalent on Twitter and Instagram, the platforms that revolutionized their use (Butwin 2016); but they are also frequently used on many other platforms, such as Facebook and Pinterest. Businesses have realized the power of hashtags as a way to steer other users toward posts about a particular topic or idea (firm-generated content) on social media.

Hashtags facilitate searches by grouping topics of interest into keyword collections. Firms use hashtags as a social media version of trademarks to promote products and services, share news, and engage with consumers on social media platforms. Several studies have investigated the effect of trademarks on firms' critical measures. For example, through trademarking in general, a firm conveys information regarding new products, services, marketing, and brand awareness efforts and intent to build trust and influence consumer behavior (Crass et al. 2019, Faurel et al. 2021, Yang and Yuan 2019). However, trademarking hashtags is a relatively new and important phenomenon. There are a few notable legal studies from a different stream of literature (Arthur 2014, Falconer 2016, Chu 2017, McKelvey and Grady 2017, p. 90). Chu (2017, p. 413) studies the trademark protection for hashtags and concludes, “Competitors, who use the hashtags to mislead consumers, pose a serious threat to the reputation of companies who have the authority to use the hashtags.” They urge practitioners of trademark law not to lag behind the curve. Similarly, Curtin (2010) stresses the need for congressional legislation to prevent abuses like cybersquatting, the practice of capitalizing on public deception by owning domain names very similar to those of large companies (e.g., citibank.org versus citybank.org). McKelvey and Grady (2017) favor strict enforcement of trademarks; but they also call attention to “the risk in stifling conversations stimulated by hashtags,” which “may call for more innovative tactics to achieve brand protection and consumer protection goals.” Another article, which uses sports as a case study, recognizes the speed with which damage can be done to a player or team brand, leading the authors to call for more consistency in the court rulings about trademarks in the social media realm (Arthur 2014). In fact, they deem social media to be inherently commercial,

representing their vote that hashtags can be trademarked just like any other word or phrase. However, firms' social media strategies will be different after an expected event—trademarking a hashtag. We fill this important gap by extending the prior literature on trademarks in a social media context: investigating the signaling effect of trademarking a hashtag on a firm's social media audience engagement.

3. Data Collection, Data Description, and Summary Statistics

We have collected data from multiple sources, processed it further, and derived key features to use in our empirical analysis. In this section, we explain the data collection process and data description in detail.

3.1. Data Collection

The USPTO maintains an extensive database of trademarks dating back to 1884. This data set stores detailed information on millions of trademark applications and registrations of individuals, groups, and businesses seeking intellectual property protection. More specifically, it maintains information on trademark contents, ownership, classification, date of filing, registration, renewal or abandonment, the name of examining attorneys who examine the trademark applications, and so on.

We use the Trademark Electronic Search System supported by the USPTO. We search for the keywords “hashtag” or the pound sign symbol “#” in the entire trademark data sets for potential matches. We retrieve the following information for each trademark: serial number, registration number, year of registration, owner name, trademark description, and status (live or canceled). We process these data and include only trademarks that are marked as “registered” in the period from August 2013 to December 2017. Applications under review or rejected are excluded from our data construction as well. Also, we exclude trademarked hashtags that were assigned to individual applicants instead of to firms.

We complement our treated firms (that trademark their hashtags) with similar control firms (that do not trademark their hashtags). To find these similar firms, we first use Google's “similar” feature, which appears in each search result. This feature recommends web pages or companies that are similar to the search result one has chosen (in our case, the website of one of our treated firms), which makes control and treated firms belong to the same industry. We then manually verify that the website of this similar firm recommended using this feature is indeed similar to our focal firm, meaning that it offers similar products or services. We perform this step as an extra level of verification for our control group.

We manually identify the website of each firm that has registered a hashtag. From the firm's website, we identify the Twitter account of each firm. We extract data from each firm's official Twitter account before and after the approval of trademarking hashtags. In our study, we focus on firms' social media campaigns: hashtags created by firms' official Twitter accounts. We harvest each firm's tweets, likes received per tweet, comments received per tweet, and retweets received per tweet. Note that we do not make a distinction in terms of who retweeted a specific tweet and use a count of all the retweets posted against a particular tweet in this study. We further process the tweets by removing special characters, links, punctuation, stop words, and leading and lagging blank spaces. In total, we have 163,655 tweets in our data set. We then combine firm trademarked hashtag level information with the firm-level tweet data. Our data set consists of firms at the intersection of the trademark database and being active on Twitter.

3.2. Data Description

We have 27 treated firms and 75 control firms in our study. We have used data of the 12-month period before and after a firm trademarked a hashtag. The firms included in our study vary greatly by industry type. They include beauty/cosmetic products (L'Oréal USA), apparel (Hanna Andersson), fast food (Charleys), furniture (Crate and Barrel), pharmaceutical products (GenexaHealth), consulting (Chasse Consulting), food and beverage (West Pak Avocado), nonprofits for social goods (Girls Inc. of the Pacific Northwest, JugLife, Marine Corps Marathon Organization, Champions for Kids), automotive parts, jewelry/charms (RembrandtCharms and Monica Rich Kosann), and sports products (Taylor Made and Powersports). The inclusion of firms from multiple industries in our data set enhances the generalizability of our findings. Next, we describe the features we have developed and used in our study.

3.2.1. Features Development. Using Linguistic Inquiry and Word Count (LIWC), a computerized text analysis tool (Pennebaker et al. 2015), we operationalize the psychological construct we have defined in Section 1.2 and develop the following features: Tone, Analytic, and Clout. In LIWC, these summary variables are derived from findings and measures in the previous linguistic and psychology literature (Cohn et al. 2004, Kacewicz et al. 2014, Pennebaker et al. 2014). LIWC is a state-of-the-art psycholinguistic package that relies on an internal, default dictionary that defines which words should be counted in the target text files. It can be used to analyze the linguistic features of texts by calculating percentages of words that belong to different word categories, each category reflecting different

language and psycholinguistic dimensions of a written text. These categories include linguistic processes (such as function words' usage and verb tenses), affective processes (such as positive words and negative words), cognitive processes (such as tentative, inclusive, and exclusive words), and many other words categories. LIWC has been widely used in various linguistic and psychological studies (Tausczik and Pennebaker 2010). LIWC is particularly helpful in our Twitter setting because it incorporates informal language that is common in Twitter and Facebook posts. For example, “:)” is coded as a positive emotion word. We describe the LIWC linguistic-based summary variables we use in this study.

Tone: We operationalize another psychological construct emotionally charged style with the help of a LIWC summary variable—Tone. The tone reflects a person's optimism or pessimism using positive and negative emotion words (Cohn et al. 2004). The range of the variable Tone varies from 0–100. Using emotion-based dictionaries, LIWC assigns a low Tone score to the text with greater use of emotionally negative charged words. On the other hand, it assigns a high score to the text with greater use of emotionally positively charged words.

Analytic: We operationalize another psychological construct, narrative style, using another LIWC-based summary variable—Analytic. Analytic measures whether people write in an analytic, logical, or more intuitive and narrative style. Analytical thinking is indicated by the greater use of nouns, articles, and prepositions. Low analytic thinking (or narrative, informal thinking) is associated with fewer pronouns, auxiliary verbs, and common adverbs (Jurafsky et al. 2009, Pennebaker et al. 2014). LIWC Analytic scores range from 0–100 with higher scores indicating more analytic thinking and lower scores indicating more narrative thinking (Pennebaker et al. 2014).

Clout: Using LIWC, we quantitatively operationalize inclusive writing with confidence, leadership, and authoritative style with a summary variable called Clout (Pennebaker et al. 2015). The summary variable Clout is derived from several past research studies that relate the use of pronouns as a symbol of rank within the social hierarchy (Kacewicz et al. 2014). Its score (i.e., standardized scores that have been converted to percent) ranges from 0–100. A higher Clout score represents a highly confident writing style with more usage of we-words (e.g., we, our) and social words (e.g., friend, ally, group). On the other hand, a low Clout score represents a more uncertain and doubtful writing style with more usage of I-words (e.g., my, mine) and swear words (Kacewicz et al. 2014).

It is worth noting that constructing the variables Analytic and Clout are mainly based on the count of

function words, which is different from the conventional text analysis approach where only content words and keywords are used to analyze texts (Lee et al. 2015).

3.2.2. Text-Based Control Variables. To control for unobserved heterogeneity due to text-based key characteristics, following Aleti et al. (2019), we control for the following variables: $\log(WC)$, the log mean number of word count per tweet; $\log(WPS)$, the log mean number of words per sentence; and *Sixltr*, the mean percent of words in the text that are longer than six letters.

3.2.3. Summary Statistics. Our econometric identification relies on our panel data (each firm can be observed in multiple time periods, the unit of our analysis is a firm-month combination). We construct a panel data set including each firm's monthly number of tweets, monthly average of likes received per tweet, monthly average of comments received per tweet, and monthly average of retweets received per tweet. The total number of observations in the panel data are 2,062. Note that our study has 102 (27 treated and 75 control) firms. Because hashtags were trademarked at different times, we collected our data 12 months before and after a firm trademarked a hashtag to ensure that we have enough time periods before and after the treatment.⁸

In Table 1, we report the summary statistics of the key variables used in this study. It shows the aggregated number of retweets, likes, comments, word count, words per sentence, percentage of long words in the text, tone, analytic, and clout score for each firm (by month level) used in this study. The number of retweets received may vary by firm. The higher the number of retweets per firm tweet, the higher the level of social media engagement. The log mean number of retweets received by a firm tweet (by month) is 1.320. The log mean number of likes received by a firm is 1.762, whereas the log mean number of comments received by a firm is 0.381. Higher analytic scores indicate a more logic-based thinking style used in tweets. The average analytic score of a firm in our data set is 80.439. A higher clout score represents a more confident writing style of users. The average clout score of a firm in our data set, based on tweet content, is 72.783. A higher emotional tone (Tone) score reflects a person's optimism, whereas a low tone score reflects pessimism. The average tone score for a firm in our data set is 60.759. The log mean number of word count, words per sentence, and percent of long words in the tweet text are 2.859, 2.432, and 23.470, respectively.

Table 1. Summary Statistics of Tweets Posted by Firms

Variables	Min	Mean	Max	SD
The log mean number of retweets received per firm tweet (by month)	0	1.320	9.136	1.279
The log mean number of likes received per firm tweet (by month)	0	1.762	9.998	1.544
The log mean number of comments received per firm tweet (by month)	0	0.381	5.873	0.613
The log mean number of word count per tweet	0	2.859	4.389	0.299
The log mean number of words per sentence	0	2.432	4.317	0.339
The mean percent of words in the text that are longer than six letters	0	23.470	100	9.065
Tone	0	60.759	99	19.990
Analytic	0	80.439	99	14.025
Clout	0	72.783	99	13.466

Note. SD, standard deviation.

4. Empirical Model and Main Results

In this section, we examine our first research question: The impact of trademarking hashtags on social media audience engagement. Then, we provide various robustness checks to validate the main effect.

4.1. Impact of Trademarking Hashtags

We first present our baseline fixed effects model. Then, we address different endogeneity mechanisms of self-selected hashtag trademarking.

4.1.1. Baseline Empirical Model. Following Zhang and Zhu (2011), we examine the change in the number of retweets after trademarking hashtags in the following benchmark regression model:

$$\begin{aligned} \log(\text{Retweet}_{it}) = & c_i + \beta_0 + \beta_1 \text{Trademark}_{it} \\ & + \beta_2 \log(\text{Tweets}_{it}) \\ & + \beta_3 \text{TweetsLinguisticControls}_{it} \\ & + \beta_4 \text{OtherControls} + \varepsilon_{it}, \end{aligned} \quad (1)$$

where the dependent variable, $\log(\text{Retweet}_{it})$, is the log mean number of retweets for firm i 's posts at time t (each time period is a month), and Trademark_{it} is a dummy variable indicating whether firm i has trademarked hashtag at time t (one indicates the posttrademark period, zero indicates the pretrademark period; Trademark_{it} is always zero for firms that do not trademark hashtags), and c_i is the unobserved firm fixed effect. In this fixed effects model, we are able to control for the unobserved time-invariant firm-level heterogeneity, such as firms' size, the year established, and the country (Wooldridge 2002).

We control for three types of variables. The first type is the log number of tweets posted by firm i at time t , that is, $\log(\text{Tweets}_{it})$. The second type includes the variables describing the linguistic features of tweets posted by firms, which are $\log(\text{WC})$ (the log mean number of word count), $\log(\text{WPS})$ (the log mean number of words per sentence), Sixltr (the mean percent of words in the text that are longer than six letters), and three summary language variables (the

mean *Tone*, *Analytic*, and *Clout* scores). The third type, *OtherControls*, includes time dummies, which are controlled in every regression equation. In particular, when the independent variables refer to the numbers, we take logs. The reason is that log can be interpreted as percentage changes, and we are more interested in the impact of a percentage increase than the impact of an increase in the number (Wooldridge 2015).

The estimation results are presented in columns (1) and (2) of Table 2 (column (1): without controls; column (2): with controls). We are interested in the coefficient of Trademark_{it} , and it is significantly positive in columns (1) and (2). The implication is that trademarking a hashtag can increase the number of retweets of a firm by about 27%, which is a considerable amount. To alleviate concerns about the failure to meet standard regression assumptions, such as clustering and heteroskedasticity, we use the robust statistics clustered at the firm level.

We also find that the coefficient of $\log(\text{Tweets})$ is positive and statistically significant, which suggests that the number of tweets posted by a firm is positively associated with the retweeting behavior. This result is consistent with the prior research that a firm's social media posts can increase engagement in online brand communities (Bapna et al. 2019). The coefficients of *Tone* and *Clout* indicate that tweets written in a more positive tone or with a more confident style are more likely to be retweeted. However, the coefficient of *Analytic* shows that tweets written in a less analytic style are more likely to be retweeted. In other words, people tend to like a more narrative and informal writing style in tweets. The coefficients of *Tone*, *Analytic*, and *Clout* are not only statistically significant but also practically significant. For instance, a standard-deviation increase in *Tone* can be associated with a 6.2% increase in retweets ($0.31\% \times 19.99 = 6.2\%$). In our context, firms could use social media posts as a means of influencing the perceptions of consumers and increase social media engagement (Zott and Huy 2007). Our results imply that social media engagement can be enhanced through social media posts with a more positive tone, in a more confident

Table 2. The Impact of Trademarking Hashtags on Social Media Consumer Engagement

Variables	FE Robust S.E. (1)	FE Robust S.E. (2)
Trademark	0.273*** [2.638]	0.274** [2.576]
<i>Log(Tweets)</i>		0.0838** [2.045]
<i>Log(WC)</i>		−0.153 [−0.928]
<i>Log(WPS)</i>		0.181 [1.379]
<i>Sixltr</i>		0.000193 [0.0468]
<i>Tone</i>		0.00310** [2.217]
<i>Analytic</i>		−0.00347*** [−3.682]
<i>Clout</i>		0.00283** [2.149]
Time dummies	Yes	Yes
Constant	1.247*** [6.711]	1.645*** [2.890]
Observations	2062	2062

Notes. Robust t-statistics in brackets. FE, Fixed effects; SE, Standard errors; DV, Dependent variable.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

style, and in a more intuitive and narrative writing style (less analytic style of writing).

We address the issue of multicollinearity in regression equations by looking at the variance inflation factors (VIFs). The VIF quantifies the severity of multicollinearity in a regression. When the independent variables are correlated, the estimated standard errors of the fitted coefficients are inflated, which can be used to check for the presence of multicollinearity (Chatterjee and Hadi 2012). In the absence of any linear relationship among independent variable j and other independent variables, VIF_j would be one. The deviation of the VIF_j value from one indicates departure from orthogonality and tendency toward collinearity. VIFs are widely used in the information systems literature to establish if multicollinearity is an issue (e.g., Burtch et al. 2013, Singh et al. 2014, Bapna et al. 2019, Khurana et al. 2019). Most prior studies rely on informal rules of thumb applied to the VIF. For instance, in Chatterjee and Hadi (2012, p. 250), “values of variance inflation factors greater than 10 are often taken as a signal that the data have collinearity problems.” We find that the VIFs associated with each variable are less than two, which is well below the conventionally accepted threshold ($VIF < 10$), indicating that multicollinearity is not an issue in our model and does not significantly impact our findings.

As a robustness check, we rerun our regression model (1) with two new dependent variables: the log mean number of likes received by firm i at time t , $\log(Likes_{it})$, and the log mean number of replies

received by firm i at time t , $\log(Reply_{it})$. We use these two variables as additional performance measures for a firm’s social media engagement. The estimation results are robust and are presented in Table 3.

Note that we use firm-month as our unit of analysis instead of firm-hashtag to avoid a data sparsity issue. Data sparsity is an issue in which a relatively high percentage of a variable or a set of variables do not contain actual data. In our case, many hashtags posted by firms may have been tweets or liked in a particular month. In other words, if we construct the data using firm-hashtag as our unit of analysis, the vast majority of observations might be zero, which is an inefficient rare event data set pointed out by King and Zeng (2001). From an estimation perspective, because of data sparsity issues, the estimated effects are usually biased in nature (King and Zeng 2001, Greenland et al. 2016).

Prior literature suggests several techniques, including data aggregation methods, to overcome or mitigate biased estimates issues due to data sparseness. For example, Hwangbo and Kim (2017) use several aggregation functions, such as average and sum, in recommendation systems settings where data sparsity is a major challenge. In another study, Bhowmik et al. (2016) show that the parameter estimation from aggregated data are more accurate than the parameter estimation obtained from nonaggregated or “individual” samples. Following the literature focusing on data sparseness, we aggregate our data set at the firm-month level in this study.

We realize that a firm can have both trademarked hashtags and nontrademarked hashtags. As an additional analysis, we look at two categories of each firm, trademarked hashtags and nontrademarked hashtags, and conduct an additional analysis using firm-category-month as our unit of analysis. Here, *category* means that each firm has two categories: trademarked hashtags and nontrademarked hashtags. In other words, for each firm and each month, we have two observations: one observation for trademarked hashtags and the other for nontrademarked hashtags. Using firm-category-month as our unit of analysis, on one hand, can provide a more granular analysis, and, on the other hand, can alleviate the data sparsity issue mentioned earlier. The results of the firm-category-month analysis are robust and are presented in Table B.2 in Online Appendix B. We also conduct an additional robustness check using weekly data. The results of the firm-week analysis are robust and are presented in Table B.3 in Online Appendix B.

We have further controlled for the contextual characteristics of the tweets. To help us identify the latent topics about which firms are tweeting, we use latent Dirichlet allocation (LDA), a type of topic modeling method based on text mining, to identify tweet topics in an automated and scalable manner (Lee et al. 2016,

Table 3. The Impact of Trademarking Hashtags on Social Media Consumer Engagement: Additional Social Media Performance Measures

Variables	(1) DV: Log(Likes), FE Robust S.E.	(2) DV: Log(Reply), FE Robust S.E.
<i>Trademark</i>	0.233*** [2.902]	0.120*** [2.772]
<i>Log(Tweets)</i>	0.0787** [2.263]	0.0450** [2.192]
<i>Log(WC)</i>	−0.174 [−1.054]	−0.104 [−1.132]
<i>Log(WPS)</i>	0.173 [1.345]	0.108 [1.396]
<i>Sixltr</i>	−0.00230 [−0.560]	−0.00331 [−1.656]
<i>Tone</i>	0.00196** [2.396]	0.000669** [2.067]
<i>Analytic</i>	−0.00162*** [−2.934]	−0.00291*** [−2.644]
<i>Clout</i>	0.00463** [2.051]	0.00101*** [2.854]
Time Dummies	Yes	Yes
Constant	1.687*** [2.680]	0.792* [1.821]
Observations	2,062	2,062

Notes. Robust t-statistics in brackets. FE, Fixed effects; SE, Standard errors; DV, Dependent variable.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Bapna et al. 2019, Pu et al. 2020, Pu et al. 2021). The input to the LDA topic model is a set of documents, which in our study is the text of firms' tweets. The output of LDA is a predefined number (K) of topics from these posts. LDA provides a posterior topic distribution over all K topics for each post. In other words, for each tweet j , the LDA model outputs a K -vector, $\langle T_{j,1}, T_{j,2}, \dots, T_{j,K} \rangle$, where $T_{j,k}$ represents the weight of topic k associated with tweet j . In addition, for each tweet j and topic k , $0 \leq T_{j,k} \leq 1$ and $\sum_{k=1}^K T_{j,k} = 1$. The larger the weight of a given topic, the more likely it is that the post is associated with that topic. We aggregate the average topic weight at the firm-month level and find consistent results after controlling for the topics generated from LDA. The results are presented in Online Appendix B. We have also collected a small data sample from Instagram and show that our results are robust on Instagram. Please refer to Online Appendix B for details.

It is worth noting that, in our specific context, the control group is the group of firms that does not trademark hashtags and the treatment group is the group of firms that does trademark hashtags. A typical endogeneity concern is that firms are self-selected to trademark hashtags. In other words, the treated firms are not randomly selected: The firms that trademark hashtags might be systematically different from the firms that do not trademark hashtags. An important source of systematic differences is that firms who trademark hashtags might be more likely to launch social-media-focused campaigns than firms who do

not trademark hashtags. In other words, firms' unobserved efforts in marketing themselves could affect their decisions to trademark hashtags. In the following subsection, we conduct various analyses to address the endogeneity concerns.

4.1.2. Causal Identification Strategies in Addressing Endogeneity Concerns. To address different endogeneity mechanisms of self-selected hashtag trademarking, we use various identification strategies. If the selection process is unknown, we have two cases with different identification assumptions: (i) the selection process is driven by observable characteristics (observable to researchers) and (ii) the selection process is driven by unobserved characteristics. In case (i), we use a combination of the fixed effects model with the propensity score matching (PSM) approach to correct the possible endogeneity (see Section 4.1.3). In case (ii), we address the endogeneity concern by using a quasi-experimental design (see Section 4.1.4), which is essentially the Look-Ahead Propensity Score Matching (LA-PSM; see Bapna et al. 2018). We validate the parallel trends assumption in Section 4.1.5.

4.1.3. Propensity Score Matching. To avoid model dependence, we also consider a fixed effects regression model combined with matching techniques, such as propensity score matching. Following Goh et al. (2013), Li (2016), Chen et al. (2021), and Pan and Qiu (2022), we first create a "proper" control group for treated firms by using PSM. We ensure that the

control and treated groups are comparable in terms of observable characteristics. Then, we run the fixed effects regression Equation (1). In this regression, for the treated firms, we use the observations before and after trademarking hashtags.

In the matching process, the treated firms are the ones that trademark hashtags. As stated, we have 27 treated firms and 75 control firms in our research study. Using PSM, we match each treated firm to the most “similar” control firm (closest propensity score, one-to-one matching) in terms of the following characteristics: (i) the number of days that firm i has joined Twitter, (ii) the number of followers/following/tweets indicated on firm i ’s Twitter account at the beginning of our sample period, and (iii) the industry the firm belongs to. Therefore, in the matched sample, we have 27 treated firms and 27 control firms.

We first sort all firms in a random order to make sure that the ordering does not affect the subsequent matching. Then, we run a logit regression based on the pretreatment variables mentioned earlier and obtain the predicted propensity scores. We use the nearest neighbor-matching algorithm in which each treated unit is matched with the control unit with the closest propensity score. In order to assess the quality of matching, we perform t -tests of equality of means before and after the matching to check whether our PSM adequately balances characteristics between the treatment and the control group units. The results are presented in Table 4.

In Table 4, there is clear evidence of covariate imbalance between groups before matching. After matching, most of the differences of mean are no longer statistically significant, suggesting that matching helps reduce the bias associated with the observable characteristics. Figure D.3 in Online Appendix D displays a graphical summary of covariate imbalance showing the standardized percentage bias for each covariate. We can see that the covariate imbalance has been greatly reduced after matching. After matching, only one characteristic, the number of following, is not completely balanced and the imbalanced degree of the number of following is reduced. Therefore, in general, we think the matching is successful. If we only use matching, the imbalanced characteristic will bias

our estimation. However, our analysis does not depend only on the matching method. After we obtain the matched sample, we use a fixed effects model to analyze the impact of trademarking hashtags. Because the number of following at the beginning of our sample period (the imbalanced characteristic) is a time-invariant variable, it has been absorbed into the fixed effects and been controlled in the fixed effects model.

Common support or overlap condition is a critical assumption in matching. The prior matching literature (Heinrich et al. 2010) suggests that checking the overlap or region of common support between treatment and control groups can be done through a visual inspection of the propensity score distributions for both the treatment and control groups. Figure D.4 in Online Appendix D graphs the propensity score histogram by treatment status, and it reveals a clear overlapping of the distributions between treatment and control groups.

Next, we reestimate our fixed effects model (1) using the new sample created by PSM; the results are presented in column (1) of Table 5. The basic findings are consistent with those in our fixed effects model. Note that the number of observations in the control and treatment groups before and after matching is different (the number of observations before matching is 2,062, and the number of observations after matching is 1,092).

If there are unobserved time-varying variables that simultaneously affect assignment into treatment and the outcome variable, PSM may not be robust against this hidden bias. The basic idea of Rosenbaum bounds is to examine if PSM is sensitive to hidden bias: We can manipulate the estimated odds of receiving a treatment to see how much the estimated treatment effects may vary. In other words, we want to determine how strongly an unmeasured variable must influence the selection process to undermine the inference of the matching analysis.

In order to estimate the extent to which such “selection on unobservables” may bias our estimation and inference, we present the results of a Rosenbaum bounds sensitivity using Wilcoxon’s signed rank test (Rosenbaum 2002) in Table 6. It reports p -values from Wilcoxon signed rank tests for the averaged treatment

Table 4. Differences in Mean Before and After Matching

Variables	Before matching				After matching			
	Mean treated	Mean control	t -statistics	p -value	Mean treated	Mean control	t -statistics	p -value
<i>Days from joining Twitter</i>	988.7	1,551.6	−2.55**	0.012	988.7	930.37	0.22	0.829
<i>Tweets</i>	988.7	11,991	−3.42***	0.000	5,777.6	5,272.7	0.16	0.875
<i>Following</i>	988.7	2,324.6	−3.67***	0.000	1,851.9	1,937.1	2.21**	0.029
<i>Followers</i>	988.7	2.7e+05	−4.25***	0.000	49,084	44,203	0.12	0.904

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table 5. The Impact of Trademarking Hashtags on Social Media Consumer Engagement: Addressing Endogeneity Concerns

Variables	Fixed effects+ PSM (1)	Fixed effects+ LA-PSM (2)
<i>Trademark</i>	0.235** [2.384]	0.228** [2.314]
<i>Log(Tweets)</i>	0.954** [2.115]	0.107** [2.327]
<i>Log(WC)</i>	−0.134 [−0.821]	−0.112 [−0.633]
<i>Log(WPS)</i>	0.127 [1.057]	0.119 [1.221]
<i>Sixltr</i>	0.000288 [0.0587]	0.000274 [0.0545]
<i>Tone</i>	0.00394*** [2.775]	0.00413*** [2.861]
<i>Analytic</i>	−0.00386*** [−3.027]	−0.00395*** [−3.144]
<i>Clout</i>	0.00295** [2.284]	0.00326** [2.335]
Time dummies	Yes	Yes
Constant	1.526*** [2.971]	1.481*** [2.902]
Observations	1,092	435

Note. Robust t-statistics in brackets.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

effect while setting the level of hidden bias to a certain value Γ . In Table 6, Γ is the odds ratio of treatment assignment, which is a measure of the degree of departure from a study that is free of hidden bias. Our sensitivity analysis considers several possible values of Γ and shows how the inferences might change. When $\Gamma = 1$, it implies that we assume the absence of unobserved selection bias. In this case, both the upper of lower bounds of p -values are zero ($\text{sig+} = 0$, $\text{sig-} = 0$) in Table 6 indicating that the effect of trademarking hashtags is significant when there is no hidden bias. We interpret the p -values under different values of Γ in Table 6 as follows: PSM is sensitive to hidden bias if values of Γ close to one could lead to inferences that are very different from those obtained assuming the study is free of hidden bias ($\Gamma = 1$). PSM is insensitive if extreme values of Γ are required to alter the inference.

The critical level of Γ at which we would have to question the significance of our PSM is between four and five (when $\Gamma = 5$, the upper bound of p -value is greater than 10%): For the impact of trademarking hashtags to disappear, the unobserved confounder has to cause the odds ratio of treatment assignment to differ between treatment and control groups by a factor of about five. Note that a Γ value of five is very large (Guo and Fraser 2010), and it is well above the threshold values used in the prior studies for robust PSM (e.g., Wei and Lin 2017). It is also worth noting that the Rosenbaum bounds are “worst-case” scenarios. An insignificant upper bound p -value for $\Gamma = 5$ does not mean that there is no true positive effect of

Table 6. Rosenbaum Bounds for PSM: Range of Significant Levels for the Signed Rank Statistic

Γ	Sig+	Sig−
1	0	0
2	0	0
3	0.000445	0
4	0.0836	0
5	0.482	0

Note. Γ , odds of differential assignment due to unobserved factors; Sig+, upper bound significance level; Sig−, lower bound significance level.

trademarking hashtags when $\Gamma = 5$. This result means that the confidence interval for the effect of trademarking hashtags will include zero if an unobserved variable causes the odds ratio of treatment assignment to differ between treatment and control groups by five. As a summary, the results in Table 6 show that our PSM is robust to a plausible range of unobserved selection bias (hidden bias).

4.1.4. A Quasi-Experimental Design (Fixed Effects Model Combined with LA-PSM).

The gold standard of establishing convincing causality is to run randomized experiments. In our context, an ideal experiment is to randomly assign firms into a control or treatment group (whether they trademark hashtags or not). However, randomized trials are not always available because they are expensive. Following similar identification ideas to those in Bapna et al. (2018) and Wang et al. (2018), we use a quasi-experimental design to account for the differences between control and treatment groups caused by unobserved characteristics.

The intuition of our quasi-experiment research design is to exploit the time sequence of firms’ decisions on trademarking hashtags and identify a more similar control group in terms of unobserved characteristics. For example, an econometric challenge in Garg et al. (2011) and Wang et al. (2018) is to separate causal peer influence from homophily (the tendency of individuals to associate with similar others). Empirically, it is not difficult to find that a focal user’s behavior is more likely to be correlated with his or her online friends’ behaviors than with anonymous users. However, this empirical finding can be driven by either causal peer influence or homophily. The difficulty of establishing causal inference of peer influence is a result of lacking a proper control group. A focal user’s friends (treatment group) can be very different from anonymous users (control group) in many characteristics.

To ensure similarity between control and treatment groups, Wang et al. (2018) identify a proper control group by observing new social ties formed in a future time period: A proper control unit is a person who is

not the focal user's online friend now but will be his or her online friend at a future point. This quasi-experiment research design ensures that the treated units and the control units are similar in unobserved characteristics. Their identification intuition also applies to our context. Firms that trademark hashtags (treated units) can be very different from those that do not trademark hashtags (control units) in terms of observable and unobserved characteristics. We can use the same approach as in Bapna et al. (2018) and Wang et al. (2018) to identify a better control group. Basically, a better control unit could be a firm that has not trademarked hashtags but will trademark hashtags in the future. Similarly, this quasi-experiment design in our context ensures that the treated firms and the control firms are similar in unobserved characteristics.

To account for both observable and unobserved characteristics as best as we can, we adopt the LA-PSM method proposed by Bapna et al. (2018) to identify the proper control group and then we combine it with the fixed effects model. More specifically, we first split the whole sample period into two equally long time periods (period I and period II). The treatment group in our quasi-experimental design is the firms that trademark hashtags in time period I. For each treated unit, we match it to a control unit with the closest propensity score among firms that have not trademarked hashtags in time period I but will trademark hashtags in time period II. On the one hand, the closest propensity score ensures that the treated and the control units are similar in observable characteristics (the observable characteristics in PSM are the same as those in Section 4.1.3); on the other hand, choosing firms from those that have not trademarked hashtags in time period I but will trademark hashtags in time period II ensures that the treated and the control units are similar in unobserved characteristics.

After creating the proper control group, we construct a new data sample with treated and control units in time period I. We reestimate our fixed effects regression Equation (1) using the new data sample. The results are presented in column (2) of Table 4 and are consistent with those in the PSM model. The differences in mean before and after LA-PSM is presented in Online Appendix B.

4.1.5. Additional Robustness Checks. If time-variant unobserved confounders cause a significant heterogeneity in the pretreatment trends, it suggests that the pretreatments may disproportionately affect treated units, as opposed to control units and the “parallel path” assumption is less likely to be satisfied (Angrist and Pischke 2008). In our context, the concern of pretreatment trends arises because firms' unobserved efforts in marketing themselves could affect their decisions to trademark hashtags. In addition, there might

be reverse causality concerns: A firm might be more likely to trademark hashtags because of the increased level of social media engagement. In Online Appendix A, we adopt the relative time model proposed by Autor (2003) to address the concerns of reverse causality and pretreatment trends.

We conduct a placebo test to examine if our results could be driven entirely by chance. The idea of the placebo test proposed here is akin to the framework in Bertrand et al. (2004): We estimate pseudocausal effects that are known to be equal to zero based on a priori knowledge. More specifically, following Bertrand et al. (2004), we randomly generate placebo trademarked hashtags, where treated firms are chosen at random. Because these trademarked hashtags are fictitious, a significant effect of trademarking hashtags at the 5% level (5% significance level) should be found roughly 5% of the time. If in the placebo test, a significant effect at the 5% level is found at a value much larger than 5% of the time, then our interpretation is that our analysis could be driven by the placebo effect and does not provide significant evidence of a positive effect of trademarking hashtags. On the other hand, if a significant effect at the 5% level is found at a value about 5%, then our interpretation is that our analysis provides evidence that the positive effect of trademarking hashtags is not driven by the placebo effect.

Following Bertrand et al. (2004), we reestimate regression Equation (1) using these placebo trademarked hashtags. The estimation generates an estimate of the effect of trademarking hashtags and a standard error for this estimate. We repeat this exercise 1,000 times. Because these trademarked hashtags are fictitious, we expect to reject the null hypothesis of no effect roughly 5% of the time (50 times). From our results for the 1,000 runs, we find that the fraction of simulations in which the null hypothesis is rejected is 3.2% (32 times), indicating that our results are unlikely to be driven by the placebo effect.

Additionally, we conduct another robustness check including firm-specific time trends by estimating a correlated random trend model (Angrist and Pischke 2008). The results are robust and are presented in Online Appendix A.

4.2. Heterogeneous Effects of Trademarking Hashtags

In this section, we examine how the effect of trademarking hashtags is moderated by the number of Twitter followers and linguistic features of writing tweets. Then, we investigate the contemporaneous and lagged effects of writing tweets with desirable linguistic styles in trademarking hashtags.

4.2.1. The Moderating Effects of Twitter Followers. First, we investigate our second research question: whether

Table 7. The Impact of Trademarking Hashtags on Social Media Consumer Engagement: Moderating Analysis

Variables	log (Retweet) (1)
Trademark	0.353*** [2.729]
Trademark*TwitterFollower	−0.224** [2.084]
Log(Tweets)	0.0732** [2.154]
Log(WC)	−0.143 [−0.658]
Log(WPS)	0.153 [1.165]
Sixltr	0.000167 [0.0283]
Tone	0.00328*** [3.025]
Analytic	−0.00389*** [−3.427]
Clout	0.00261** [2.175]
Time dummies	Yes
Constant	1.421*** [4.275]

Note. Robust t-statistics in brackets.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

the magnitude of the effect of trademarking hashtags differs among firms with different levels of Twitter followers. More specifically, we use a dummy variable to indicate a high/low level of Twitter followers (*TwitterFollower* = 1 if in the top 50 percentile level of Twitter followers, *TwitterFollower* = 0 otherwise) and add an interaction term into our regression model.

The estimation results are presented in Table 7. We find that the coefficient of the interaction term is significantly negative, which suggests that the positive effect of trademarking hashtags is stronger for firms with fewer Twitter followers. As mentioned earlier, in the signaling theory, the strength of signals depends on the level of information asymmetry: Signals have more informational value (the effect of signaling becomes larger) when the information asymmetry is higher (Hottenrott et al. 2016). In our context, consumers might not be familiar with firms with fewer Twitter followers, including their products and services, because of high information asymmetry. Therefore, we observe that the positive effect of signaling through trademarking a hashtag is larger for firms with fewer Twitter followers. Essentially, the smaller, less-established firms use trademarking hashtags to signal and legitimize their credibility. We also conduct a subsample analysis on Twitter followers in Online Appendix B, and the results are robust.

4.2.2. The Role of Linguistic Features of Writing Tweets in Trademarking Hashtags. In this section, we examine our third research question and focus on the

Table 8. The Moderating Impact of Trademarking Hashtags

Variables	FE robust S.E. (1)
Trademark	0.149** [2.287]
Log(Tweets)	0.0531** [2.026]
Log(WC)	−0.173 [−1.160]
Log(WPS)	0.190 [1.523]
Sixltr	0.000850 [0.236]
Tone	0.00184** [2.209]
Analytic	−0.00264** [−2.418]
Clout	0.00168** [2.220]
Log(Tweets)*Trademark	0.218*** [3.390]
Tone*Trademark	0.00534*** [3.249]
Analytic*Trademark	−0.00585*** [−4.661]
Clout*Trademark	0.00461*** [3.654]
Time dummies	Yes
Constant	1.624*** [3.122]
Observations	2,062

Notes. Robust t-statistics in brackets. FE, Fixed effects; SE, Standard errors; DV, Dependent variable.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

key underlying mechanisms. More specifically, we ask if trademarking hashtags can make firms' tweets with certain linguistic features more effective in increasing social media audience engagement. We estimate the following regression equation using the base control/treatment sample:

$$\begin{aligned} \log(\text{Retweet}_{it}) = & c_i + \beta_0 + \beta_1 \text{Trademark}_{it} + \beta_2 \log(\text{Tweets}_{it}) \\ & + \beta_3 \text{TweetsLinguisticControls}_{it} \\ & + \beta_4 [\log(\text{Tweets}_{it}) \cdot \text{Trademark}_{it}] \\ & + \beta_5 (\text{Tone}_{it} \cdot \text{Trademark}_{it}) \\ & + \beta_6 (\text{Analytic}_{it} \cdot \text{Trademark}_{it}) \\ & + \beta_7 (\text{Clout}_{it} \cdot \text{Trademark}_{it}) \\ & + \beta_8 \text{OtherControls} + \varepsilon_{it}. \end{aligned} \quad (2)$$

The estimation results are presented in Table 8. First, it indicates that without trademarking hashtags, a percent increase in the number of tweets is associated with about 0.05% increase in retweeting behavior; with trademarking hashtags, a percent increase in the number of tweets is associated with about 0.27% ($0.22 + 0.05 = 0.27\%$) increase in retweeting.

Table 9. The Impact of Trademarking Hashtags: Contemporaneous Effects vs. Lagged Effects

Variables	FE robust S.E. (1)
<i>Trademark</i>	0.172** [2.227]
<i>Log(Tweets)</i>	0.0311** [2.017]
<i>Log(WC)</i>	−0.133 [−0.685]
<i>Log(WPS)</i>	0.233 [1.644]
<i>Sixltr</i>	0.00121 [0.244]
<i>Tone</i>	0.00124** [2.138]
<i>Analytic</i>	−0.00242** [−2.126]
<i>Clout</i>	0.00147** [2.238]
<i>Lag.Log(Tweets)</i>	0.0301 [0.782]
<i>Lag.Log(WC)</i>	−0.171 [−0.780]
<i>Lag.log(WPS)</i>	−0.0234 [−0.173]
<i>Lag.Sixltr</i>	−0.000191 [−0.0542]
<i>Lag.Tone</i>	0.00199 [1.426]
<i>Lag.Analytic</i>	−0.000508 [−0.228]
<i>Lag.Clout</i>	0.00182 [1.051]
<i>Log(Tweets)*Trademark</i>	0.197*** [2.971]
<i>Tone*Trademark</i>	0.00410*** [3.121]
<i>Analytic*Trademark</i>	−0.00351*** [−3.408]
<i>Clout*Trademark</i>	0.00263*** [3.197]
<i>Lag.Log(Tweets)*Trademark</i>	0.115** [2.176]
<i>Lag.Tone*Trademark</i>	0.00223** [−2.079]
<i>Lag.Analytic*Trademark</i>	−0.00128** [2.155]
<i>Lag.Clout*Trademark</i>	0.00110** [2.232]
Time dummies	Yes
Constant	1.969*** [2.816]
Observations	1,960

Notes. Robust t-statistics in brackets. FE, Fixed effects; SE, Standard errors; DV, Dependent variable

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

It is worth noting that although the coefficient of the interaction terms with log number of tweets is larger than the coefficients of all other interaction terms combined, it does not mean that the log number of tweets matters more than everything else combined. The reason is that these independent variables have different

units: (i) For log number of tweets (a log-log model), the meaning of coefficient on the interaction term (0.218) is that after trademarking hashtags, a percent increase in the number of tweets is associated with an additional 0.218% increase in the number of retweets. (ii) For Tone (a log-level model), the meaning of coefficient on the interaction term (0.00534) is that after trademarking hashtags, a unit increase in the Tone score (ranges from 0–100) is associated with an additional 0.534% (100×0.00534 , because it is a log-level model) increase in the number of retweets. (iii) For Clout (a log-level model), the meaning of coefficient on the interaction term (0.00461) is that after trademarking hashtags, a unit increase in the Clout score (ranges from 0–100) is associated with an additional 0.461% (100×0.00461 , because it is a log-level model) increase in the number of retweets. (iv) For Analytic (a log-level model), the meaning of coefficient on the interaction term (−0.00585) is that after trademarking hashtags, a unit increase in the Analytic score (ranges from 0–100) is associated with an additional 0.585% (100×0.00585 , because it is a log-level model) decrease in the number of retweets. Therefore, the number of tweets does not seem to play a more important role than anything else combined.

We find that the coefficients of the interaction terms *Tone*Trademark* and *Clout*Trademark* are positive and statistically significant, and these two coefficients are much larger than those on *Tone* and *Clout*, respectively ($0.00534 > 0.00184$, $0.00461 > 0.00168$). Our findings show that without trademarking a hashtag, a standard-deviation increase in “tone, clout” scores is associated with a 3.68% ($= 0.00184 \times 19.99$) or 2.26% ($= 0.00168 \times 13.47$) increase in retweeting, respectively. However, trademarking hashtags will make the magnitude much larger: A standard-deviation increase in “tone, clout” scores is associated with a 14.35% ($= (0.00534 + 0.00184) \times 19.99$) or 8.47% ($= (0.00461 + 0.00168) \times 13.47$) increase in retweeting, respectively. Finally, we find that the coefficient of the interaction term *Analytic*Trademark* is negative and statistically significant, and its absolute magnitude is much larger than that on *Analytic* ($|-0.00585| > |-0.00264|$). The implication is that without trademarking a hashtag, a standard-deviation decrease in analytic scores is associated with a 3.70% ($= -0.00264 \times 14.03$) increase in retweeting. However, trademarking hashtags will make the magnitude much larger: A standard-deviation decrease in analytic scores is associated with an 11.91% ($= (-0.00585 - 0.00264) \times 14.03$) increase in retweeting.

4.2.3. Contemporaneous Effects vs. Lagged Effects of Linguistic Styles in Trademarking Hashtags. In this section, we examine our fourth research question: the contemporaneous and lagged effects of writing tweets with desirable linguistic styles in trademarking hashtags.

We estimate the following regression model:

$$\begin{aligned} \log(\text{Retweet}_{it}) = & c_i + \beta_0 + \beta_1 \text{Trademark}_{it} + \beta_2 \log(\text{Tweets}_{it}) \\ & + \beta_3 \text{TweetsLinguisticControls}_{it} \\ & + \beta_4 \text{TweetsLinguisticControls}_{it-1} \\ & + \beta_5 [\log(\text{Tweets}_{it}) \cdot \text{Trademark}_{it}] \\ & + \beta_6 (\text{Tone}_{it} \cdot \text{Trademark}_{it}) \\ & + \beta_7 (\text{Analytic}_{it} \cdot \text{Trademark}_{it}) \\ & + \beta_8 (\text{Clout}_{it} \cdot \text{Trademark}_{it}) \\ & + \beta_9 \log(\text{Tweets}_{it-1}) + \beta_{10} [\log(\text{Tweets}_{it-1}) \\ & \cdot \text{Trademark}_{it}] + \beta_{11} (\text{Tone}_{it-1} \cdot \text{Trademark}_{it}) \\ & + \beta_{12} (\text{Analytic}_{it-1} \cdot \text{Trademark}_{it}) \\ & + \beta_{13} (\text{Clout}_{it-1} \cdot \text{Trademark}_{it}) \\ & + \beta_{14} \text{OtherControls} + \varepsilon_{it} \end{aligned} \quad (3)$$

The results are presented in Table 9. We find that the coefficients of the lagged terms of $\log(\text{Tweets})$, Tone , Analytic , and Clout are not statistically significant, indicating that when hashtags have not been trademarked, the lagged effect is not significantly controlling for the contemporaneous effects. However, the coefficients of the lagged terms of the interaction terms are statistically significant, which suggests that after trademarking hashtags, the lagged effects become important.

Our findings show that before trademarking hashtags, writing more tweets with desirable linguistic styles has only a contemporaneous effect: One month later (a time period in our model is a month), the lagged effect is not significant anymore. Trademarking hashtags makes things different. When hashtags are trademarked, the effect of writing more tweets with desirable linguistic styles lasts longer. In terms of the magnitude, the lagged effect is not as strong as the contemporaneous effect but is still considerable: about 40%–60% of the magnitude of the contemporaneous effect.

5. Discussion and Managerial Implications

In this study, we contribute to the literature on FGCs, signaling theory, and information asymmetry phenomenon by conducting research surrounded with the act of trademarking hashtags, which has not been fully explored. We believe that this lack thereof has produced an important gap in the literature and hence provides a unique opportunity for further empirical investigation. Our study differs from the broad literature dealing with UGC and FGC. Drawing on the signaling theory, we examine a specific type of firm-side (FGC) social media signaling strategy—trademarking a hashtag. We provide a theoretical framework (with

four important research gaps) that highlights how a firm's use of social media to communicate with consumers (through the lens of speech act theory) impacts their social media effectiveness after an expected signaling event—trademarking a hashtag. With the act of trademarking a hashtag, a firm sends credible signals to consumers about its seriousness to protect its brand and maintain brand exclusivity. Our study provides several important implications for businesses that are considering joining the hashtag-trademarking bandwagon as a social media signaling strategy.

First, we explore how a firm's credible signaling act through trademarking a hashtag is associated with engagement in online brand communities. We find that trademarking a hashtag is, in fact, effective in increasing social media engagement in online brand communities. Our results show that trademarking a hashtag can increase the number of retweets of a firm by about 27%, which is a considerable amount. Therefore, through the lens of signaling theory, we establish that the act of trademarking a hashtag helps a firm depict itself as credible by displaying its capability and commitment in building brand loyalty and thus increase audience engagement on social media. Our results are consistent with prior literature where a signaling event leads to an increase in audience intention to engage with a firm (Jeng 2016).

Using several causal identification strategies, our study is the first to provide an empirically driven, comprehensive understanding of trademarking hashtags. Not only is estimating the signaling impact of trademarking hashtags theoretically important but it also has practical implications. Trademarking a hashtag could be a resource-intensive and time-consuming endeavor. Without quantifying the benefit of trademarking hashtags, it is not clear whether firms should embrace trademarking hashtags actively. Our empirical analysis provides researchers and practitioners the tools and evidence to aid firms in their ongoing attempts to maximize the effectiveness of their social media engagement strategies.

Second, we investigate the heterogeneity of the signaling effect of trademarking a hashtag on social media, that is, whether the signal strength or magnitude of the effect of trademarking a hashtag differs among firms with different levels of Twitter followers. We find that the strength of the signal indeed depends on the level of information asymmetry. In other words, the beneficial effect of trademarking a hashtag is stronger for less-known firms with fewer Twitter followers. One possible reason is that consumers may not be familiar with small firms and their brands because of high information asymmetry. Therefore, the signals associated with trademarking a hashtag have more informational value for firms with fewer Twitter followers (Hottenrott et al. 2016). This finding

has important managerial implications and highlights that the heterogeneous effect of trademarking hashtags does exist. Our results reveal that small and less popular firms should consider signaling through trademarking hashtags as it is found to be an effective and attractive social media strategy in terms of increasing firms' social media audience engagement.

Third, using speech act theory, we find that certain psychology-based constructs (linguistic characteristics) of firm-generated tweets are more important (perlocutionary effect) after trademarking a hashtag, which is crucial for managers' social media marketing. Posting more tweets or writing tweets with certain desirable linguistic styles is not that important when hashtags have not been trademarked because the magnitude of those effects is relatively small. However, after the act of trademarking a hashtag, firms should pay close attention to their writing styles because trademarking a hashtag makes the effects of writing tweets with desirable linguistic styles much more salient. After the successful act of trademarking a hashtag, a social media strategy should focus not only on the quantity (writing more tweets) but also on the quality, that is, how firms express themselves by writing tweets with desirable linguistic styles. In particular, our results suggest that in order to increase social media audience engagement, firms should write tweets with a more positive tone and in a more confident style. Also, firms should pursue a more intuitive and narrative style (less analytic style of writing) in writing tweets. Our results reinforce the idea that a successful social media strategy is not only about writing more tweets and posting more messages. It has multiple dimensions including whether to trademark hashtags and how to write tweets. The underlying reason for the linguistic role of features on perlocution is based on speech act theory through firms' use of utterance of words (FGC in our context). In our context, firms could use posts as a means of influencing the perceptions of others. Tone, Analytic, and Clout are firms' use of actions to create a perlocutionary effect (retweets). This effect can be achieved through social media posts with a more positive tone, in a more confident style, and a more intuitive and narrative writing style (less analytic style of writing).

Our empirical analysis shows that firms should pay close attention to how they express themselves on social media, especially after a credible signaling event—in this case, trademarking a hashtag. More specifically, after trademarking a hashtag, firms adopting a specific speech act style, that is, the utterance of words with more clout (inclusive and leadership style), narrative (storytelling), and positive emotional tone will have a stronger perlocutionary effect on the audience. Therefore, trademarking a hashtag makes the effect of writing more tweets with desirable

linguistic styles (clout, narrative, and positive) larger. Our results are consistent with prior literature on communication based on a certain linguistic style and its impact on consumer behavior (Escalas 2007, Rimé 2009, Araujo et al. 2015, Kim et al. 2016).

Finally, we find that trademarking hashtags makes writing tweets with desirable linguistic styles more important as the impact lasts longer. In summary, trademarking hashtags make both the contemporaneous and lagged effects of writing tweets with desirable linguistic styles larger and more significant. Therefore, summarizing the answers to our research questions 3 and 4, we find that after trademarking a hashtag, the amplified effects of writing tweets with desirable linguistic features may not only be more intense (answer to research question 3) but may also last longer (answer to research question 4).

We believe our study is the first to empirically investigate the signaling effect of trademarking hashtags on social media audience engagement. Therefore, we provide practical insights for researchers and businesses seeking to understand the real-world effects of trademarking a hashtag. The USPTO has allowed hashtags to be trademarked since 2013, and many firms are taking advantage. In general, the firms may not be interested in investing in trademarking hashtags if they do not see the potential returns in social media effectiveness. Our findings suggest that, despite the above concerns, it may be worth investing in trademarking hashtags. We find that the signaling effect of trademarking a hashtag plays a pivotal role in social media audience engagement. In other words, firms may observe a net positive impact in social media effectiveness after trademarking a hashtag. Moreover, firms may observe an indirect effect of trademarking hashtags: Consumer-side confidence and trust may increase because consumers may feel protected from exposure to false information disseminated through social media against a focal firm. Therefore, in order to increase social media effectiveness—especially after trademarking a hashtag—firms should consider hiring public relations specialists and dedicating resources to the development of information diffusion strategies that take communication and design into account.

6. Conclusions and Future Research Directions

Our paper focuses on FGC using Twitter along with using signaling and speech act theories and investigates the impact of trademarking hashtags on a firm's social media audience engagement. This study fills an important gap in the information systems and consumer behavior literature by providing a deep understanding of a specific social media signaling strategy:

trademarking hashtags. A trademark provides a certain level of trust for consumers so they can rely on the signals conveyed as brand equity (Katz 2010). This function instills confidence that the signals are authentic, reliable, and, therefore, trustworthy. Because of this trust function, firms seek to establish and protect their brand and send credible signals to consumers on social media. With credible signals received through trademarked hashtags, consumers adjust their perception toward the firm and engage with the firm-generated content more effectively (through retweets) and build a positive relationship with the firm (Boehmer and Tandoc 2015). Overall, it increases brand awareness and reputation, reduces consumer search costs, and fosters loyalty and trust among consumers.

We empirically examine the influence this strategy has on a firm's social media performance as well as the role of linguistic styles on writing tweets. Endogeneity concerns posed by self-selected trademarked hashtags are addressed through multiple causal identification strategies. We establish a robust quantitative relationship between trademarking hashtags and the social media performance measure. It is important for any firm, regardless of its size, to send credible signals about building and protecting its brand reputation, through trademarking a hashtag, to consumers on social media. The signaling strength of trademarking a hashtag is higher for smaller firms because of a high level of information asymmetry. Because these small firms might be less known, a weak connection with consumers or any brand-tarnishing may push them out of business. In particular, we provide empirical evidence that the beneficial effect of signaling through trademarking a hashtag is stronger for less-known firms with fewer Twitter followers.

Further, we provide insights into social media engagement strategies that can amplify social media effectiveness after hashtags have been trademarked. Firms can benefit more in terms of social media effectiveness by going beyond simply trademarking hashtags. It turns out that firms can more effectively connect with their customers by adopting specific linguistic styles. For example, our findings suggest that to increase social media audience engagement, firms should consider generating tweets in a more positive tone, with a more confident style, and with a more narrative style (less analytic). These FGCs are more likely to be retweeted after trademarking a hashtag.

Next, as competition for attention among social media users rises, trending tags may become an increasingly prized commodity. However, firms may not seek trademarks if they cannot see the long-term benefits (social media effectiveness) of spending their money, time, and other resources on trademarking hashtags. Firms may be tempted to think that trademarking hashtags and subsequently using well-crafted linguistic techniques

will only bring short-term rewards. Our research suggests that this is not the case. FGC written in certain linguistic or conversational styles is very important in general. However, our results emphasize that, after trademarking a hashtag, the amplified effects of FGC with certain linguistic features will not only be more intense but also will last longer.

There are several possible extensions to our research. First, it would be interesting to measure the spillover effect that trademarking hashtags has on nonfocal firms. It may be beneficial for focal firms to understand relevant spillover effects on direct and indirect competitors because it is possible that trademarking hashtags may benefit or hurt their direct or indirect competitors. Understanding spillover effects may help focal firms devise appropriate strategies to further intensify information diffusion after a hashtag is trademarked. Second, it would be intriguing to examine the contextual characteristics of the tweets, which may provide further insights about what hashtags to trademark and what not to trademark. Third, because the firms that we use in our study belong to different industry types, we believe the selection bias from the industries is not severe. At the same time, we believe that it would be important to study the heterogeneous effects of linguistic features for different industries. In addition, because many of the firms in our data set trademarked only one hashtag, we are not able to examine the diminishing returns of the number of trademarking hashtags in the current data set. Future research examining the diminishing returns of the number of trademarking hashtags can also guide the cost-benefit analysis of trademarking hashtags.

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Endnotes

¹ See <https://twitter.com/equalman/status/533090727081689088> (last accessed November 17, 2021).

² A show-cause order may be issued by the commissioner for trademarks if an individual or firm engages in the unauthorized practice of trademark law before the USPTO. This order requires the suspected party to legitimize their activities and explain why they should not be excluded from acting on behalf of others before the

USPTO. If the response to the show-cause order is insufficient, then an exclusion order could be issued (USPTO 2021a).

³ We have consulted with a practicing attorney in trademark law, Yoana Strateva. She has provided her endorsement of our interpretations of the scrutiny process established by the trademark offices.

⁴ See *Khaled v. Bordenave* (Southern District of New York. April 29, 2019), where the plaintiff alleges that the defendant is repeatedly using hashtags (#WETHEBEST and #WETHEBESTBRANDS) to advertise its product on social media. In this case, the court found the usage of these hashtags to clearly state a claim of trademark infringement (Samuels 2021, p. 27).

⁵ See <https://www.copyright.gov/help/faq/faq-duration.html> (last accessed November 17, 2021).

⁶ See <https://www.uspto.gov/trademarks-getting-started/trademark-basics/trademark-patent-or-copyright> (last accessed November 17, 2021).

⁷ See https://www.americanbar.org/groups/intellectual_property_law/publications/landslide/2016-17/november-december/full-court-press-hashtag-trademarks (last accessed November 17, 2021).

⁸ In other words, we have $12 + 12 = 24$ time periods for each firm. Ideally, we should have $102 * 24 = 2,448$ observations (the total number of firms * the number of periods). However, some data points are missing because some firms' Twitter accounts started less than 12 months before trademarking hashtags.

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